CS 499 CAPSTONE · TEAM SENTINEL SYSTEMS

# Houston Flood-Aware Route Planner

What the project is, how it works, where it stands, and how we split the first presentation so each of us owns a part.

## The idea in 30 seconds

Houston floods fast. Flat land, lots of concrete and storms that drop several inches in an hour can put an underpass underwater in minutes. Google Maps, Waze and Apple Maps pick the fastest route and don't know a street is flooded, so they can send drivers straight into high water.

Our app is a route planner for Houston that treats flooding as part of the route. You enter where you are and where you're going. It draws the route, checks it against flood data, and warns you if the route passes near known flooding. The goal is for it to reroute you around the danger automatically.

**Who it's for:** Houston residents and commuters, and also delivery drivers and emergency responders.

## How it works

This follows our flowchart. Steps 1–5 work in the live app today. Steps 6–7 are what we build next.

Step 1

**Enter start and destination**

Typed addresses are turned into map coordinates by Nominatim.

Step 2

**Calculate the route**

OSRM returns real driving directions, drawn on the map.

Step 3

**Read flood data**

490 Hurricane Harvey high-water marks inside Houston.

Step 4

**Check the route**

Is any flood mark within 150 m of the route line?

Step 5

**Warn the driver**

Green banner means clear. Red banner shows a risk level and water depth.

Step 6 · next

**Reroute around flooding**

Compare alternative routes and pick the safest one that's still reasonable.

Step 7 · next

**Colour the route by risk**

Each stretch of road is coloured by risk. Show safety guidance if no safe route exists.

## Tech stack, in plain words

| PIECE | WHAT IT IS | WHAT IT DOES FOR US |
|---|---|---|
| **React + Vite** | JavaScript tools for building web apps | React builds the screens. Vite sets up the project and runs it while we develop. |
| **Leaflet** | An open-source map library | Draws the map, markers, flood points and the route line. |
| **Esri World Imagery** | Satellite map tiles | The satellite background, with road labels on top, limited to the Houston area. |
| **Nominatim** | Free address search from OpenStreetMap | Turns "NRG Stadium" into latitude and longitude. |
| **OSRM** | Free, open-source routing engine | Calculates the driving route between two points. It can also return alternative routes, which we need for rerouting. |
| **USGS Harvey high-water marks** | Government survey of how high the water reached in 2017 | Our flood data: 490 points with water depth, downloaded from HydroShare. |
| **Vercel** | Web hosting | Hosts the live app at sentinelsystems.vercel.app and redeploys when the code on GitHub changes. |
| **GitHub Project** | Our task board | One issue per task, each with an owner, so everyone's contribution is tracked. |

**Terms to know:** a *high-water mark* is the highest level floodwater reached at a spot. *Geocoding* means turning an address into coordinates. A *buffer* is the 150 m zone around the route that we check for flood marks.

## Where it stands

### Done

- Houston-only satellite map
- Address search and live routing
- Harvey flood-data layer, colour-coded by depth
- On-route flood detection (150 m)
- Red/green risk warning banner
- Deployed on Vercel
- Task board on GitHub, one issue per task

### Next

- Automatic reroute around flooding (the core goal)
- A risk score for each route (0–100)
- Route coloured by risk, plus a route comparison panel
- "No safe route" safety guidance screen
- Live flood data (NWS alerts, USGS gauges)
- Testing against past flood events, plus the speed (under 5 s) and ease-of-use (3 steps or fewer) checks

# Presentation: pick your part

The deck has 11 slides. Here they're split into four parts of about 3 minutes each, so everyone speaks for a similar amount of time. Write the speaker's name under each part.

## Part 1 · Why this project

~3 min

Slides 1–3

> Good for someone comfortable opening, introducing the team and telling the story.

### Title

- Introduce the team and the project name.

### Problem Statement

- Goal: a Houston route planner people will actually use.
- Flood risk shown on the map, not hidden.
- Warn when a route passes flood-prone spots, and eventually reroute around them.

### Houston Floods Fast and Often

- Harvey (2017): thousands of streets flooded, some over 12 ft deep.
- Normal GPS apps route by distance and speed only.
- Result: drivers get sent into high water with no warning.

*Opening line idea: "Everyone here has a Houston flooding story. The apps we use to get around don't know when a road is underwater."*

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Speaker: _____

## Part 2 · How it's designed

~3 min

Slides 4–5

> Good for someone who likes explaining how the pieces connect. Use the 'How it works' and 'Tech stack' sections above.

### Our Objectives

- Walk through the flowchart: input → read flood data → flood on route? → mark segments → compute route → display.
- Point out the two endings: a colour-coded route, or safety guidance.

### System Design

- Frontend: React + Vite with a Leaflet map.
- Satellite basemap limited to Houston.
- OSRM for routes, Nominatim for address search. Both are free and open.
- Flood data: 490 USGS Harvey high-water marks.

*Opening line idea: "We didn't build a map engine from scratch. We used open routing tools and put flood data on top, so flooding becomes part of the route."*

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Speaker: _____

## Part 3 ·What's built + live demo

Slides 6–7

~3–4 min

> Best with two people: one drives the demo, the other explains what's on screen.

### Work Completed So Far

- Houston-only map with satellite imagery.
- Live routing from any start to any destination.
- 490 flood marks, coloured by depth.
- On-route detection (within 150 m).
- Red/green warning banner. Deployed live.

### Live Demo

- Case 1: TSU → Galleria. No marks within 150 m, green banner.
- Case 2: TSU → Lake Forest Park. Red banner with risk level and depth.
- Say what comes next: in the finished version it reroutes automatically.

*Opening line idea: "Here's the same app with two trips: one clear, one that passes where Harvey flooded."*

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Speaker: _____

## Part 4 ·Challenges, what's next, team

Slides 8–11

~3 min

> Good for someone who can speak to the plan and task board, and lead Q&A.

### Challenges & Solutions

- No simple live flood API, so we used USGS Harvey data (via HydroShare, no login needed).
- Map markers broke after deploying, a known Leaflet + Vite issue, fixed by importing the icons directly.
- Vite handled project setup.

### Remaining Tasks & Timeline

- Now: warn-only detection.
- Next: reroute around flooding (the core goal).
- Then: colour the route by risk.
- Stretch: live USGS gauge data.
- Mention the GitHub board: every task has an owner.

### Individual Contributions + Questions

- Let each person say one line about what they did and what they own next.
- Open the floor for questions.

*Opening line idea: "Right now the app warns you. Next, it finds you a safer way. Here's who's building each piece."*

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Speaker: _____

## Questions the professor may ask

**Why use Hurricane Harvey data instead of live data?**

We couldn't find a live flood feed that was easy to plug in, and some sources needed a login. Harvey high-water marks are real, government-surveyed data, so they're a realistic stand-in while we test live feeds like USGS stream gauges and NWS flood alerts. The detection code works the same way whatever the data source.

**Why 150 metres?**

It's roughly one to two city blocks. That's close enough that a flooded spot is likely on or right beside the road, without flagging every route in Houston. It's a starting value, and we plan to tune it when we test against past flood days.

**How will rerouting work?**

OSRM can return several alternative routes. We'll give each one a flood-risk score and recommend the safest route that isn't unreasonably longer. If every option is risky, the app shows safety guidance instead, as in our flowchart.

**How will the risk score be calculated?**

The planned inputs are how close flood points are to the route, how deep the water was, and how many flood points the route passes. They combine into a score from 0 to 100, which maps to green, yellow or red.

**How will you know it's accurate?**

We'll replay routes on past heavy-rain days in Houston and check whether the app flags the roads that were actually reported flooded. That test is in the proposal and on our task board.

**Why free APIs like OSRM and Nominatim?**

They're open source and need no paid keys, which fits a student project. The public servers have usage limits, so if we need more capacity we can run our own OSRM server or switch to Mapbox.

**How is the work divided?**

Every task is a GitHub issue with an owner on our project board. Going forward, each of us commits our own work through branches and pull requests, so contributions show up individually.

## Presenting tips

- Explain in your own words. The talking points are prompts, not a script.
- Hand off by name: "Now Kailen will walk you through the system design."
- Load the demo routes before class (TSU to the Galleria, TSU to Lake Forest Park) in case the Wi-Fi is slow.
- If a question falls in someone else's part, pass it to them. It shows everyone knows their area.